



decode request

NODE X
NODE 0

Domain X
Domain 0

long request: CP>1
short request: CP=1

Worker X
Worker 0

EngineCoreProc

MetadataServerProc

bind_domain
metadata
load_kv
shard
run_attention
commit
output

domain meta

execute model

Loop

step

scheduler

choose
cp_degree
select domain
build_domain
schedule
request finished

len / KV pool

get_domain
state
allocate
virtual_blocks
map_slot
mapping
cache_and
free_slots
init_dycp
cache

Sequential Processing

CrossDP
KVCache

VirtualBlockPool

Cross-DP HCCL Group

all-gather Q / reduce score,LSE

DP0

DP1

DPX

update

create

Shared CPU Memory / DyCP Metadata

read metadata

request_id, domain_id, cp_size

virtual_block → dp_rank, local_block

slot_mapping, seq_lens, cp_rank